

Who Hunts Who?

Comparing 4 Knowledge Graph Construction Methods for Predator–Prey Relations

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Natural Language Processing

June 2026

Motivation & Task

- ▶ Goal: automatically build a **predator-prey knowledge graph** from free text.
- ▶ Source: Wikipedia pages for 18 animals across three trophic levels.
- ▶ No animal-specific hand-written rules — only general NLP signals.
- ▶ Central question: **which extraction approach yields the best trade-off between precision and recall?**

Three trophic levels

Top predators (6)

Hawk, Eagle, Barn Owl, Fox, Wolf, Weasel

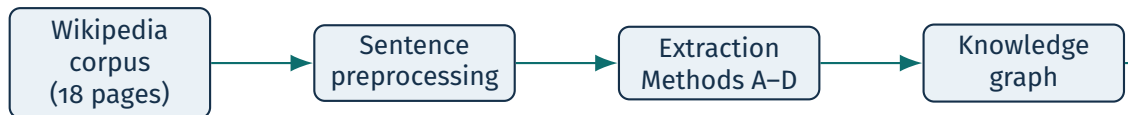
Mid-level (4)

Salamander, Lizard, Spider, Bat

Prey (8)

Mouse, Rabbit, Squirrel, Grasshopper, Cricket, Caterpillar, Fly, Beetle

Pipeline Overview



Corpus

Pages fetched via wikipedia-api, cached as JSON. Text is stripped of markup, parentheticals, and citation markers, then split into sentences.

Gold standard

38 manually curated predator→prey edges. The same benchmark is used for all four methods to ensure a fair comparison.

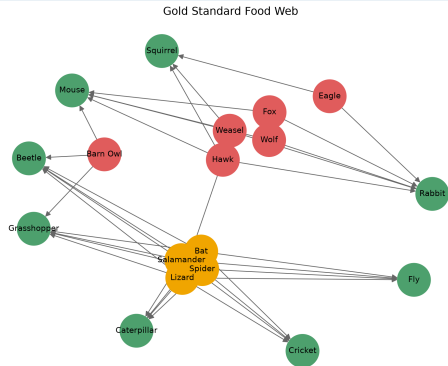
Metrics

Edge-level Precision, Recall, and F1 score. Only edges within valid trophic directions are considered.

Construction rules

- ▶ Manually constructed from Wikipedia text.
- ▶ Directed edges: predator $\rightarrow$ prey.
- ▶ Trophic constraints enforced:
 - top predators may target mid-level *and* prey.
 - mid-level predators may target prey only.
- ▶ 38 edges in total across 10 source animals.

Gold standard graph



Method A — Co-occurrence

How it works

1. For each source animal, iterate over sentences in its Wikipedia page.
2. If any target animal is mentioned in the **same sentence**: add edge source → target.
3. No verb, no syntactic analysis — pure proximity.

Example sentence:

“Foxes and rabbits both live in the forest.”

⇒ adds Fox→Rabbit

even though the sentence is about their *habitat*.

Design trade-offs

Strength: highest possible recall — every real edge that co-occurs in a sentence will be found.

Limitation: no predation signal is required. Any two animals appearing together (as competitors, taxonomic relatives, or examples) become an edge.

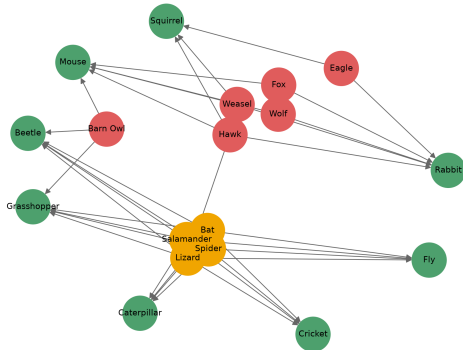
Limitation: no direction inference — edges are always assigned as source → target based on which page is being read.

Expected: very high recall, low precision.

Method A vs Gold Standard

Gold standard

Gold Standard Food Web



Method A

Method A - Co-occurrence



Scores: Precision = 0.556, Recall = 0.278, F1 = 0.370 (TP = 10, FP = 8, FN = 26)

Method B — Predation Verb Patterns

How it works

1. For each sentence containing the source animal, extract a window around both animals.
2. Requires at least one **predation verb or phrase** inside that window:
3. **Infer direction:**
 - Active: “Fox *hunts* Rabbit” → Fox→Rabbit.
 - Passive: “Rabbit *is hunted by* Fox”
Fox→Rabbit.
4. **Category expansion fallback:** “eats insects”, expands to “eats fly”

Design trade-offs

Strength: the verb requirement dramatically reduces false positives compared to Method A.

Limitation: paraphrased predation (“*diet consists of...*”, “*food source for...*”) is missed, lowering recall.

Limitation: direction inference from text position is heuristic; complex sentence structures may be mis-directed.

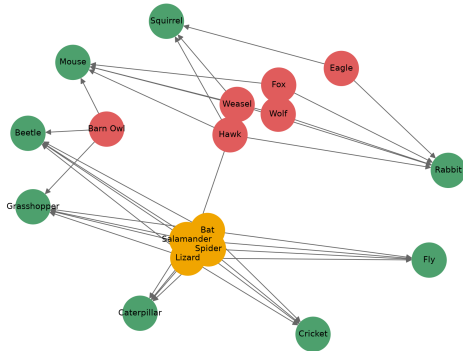
Limitation: plural forms (mice, flies) not handled uniformly in the window search.

Expected: highest precision, moderate recall.

Method B vs Gold Standard

Gold standard

Gold Standard Food Web



Method B

Method B - Dependency Patterns



Scores: Precision = 0.625, Recall = 0.139, F1 = 0.227 (TP = 5, FP = 3, FN = 31)

Method C — OpenIE-style Relation Extraction

Core idea: extract (subject, relation, object) triples.

Step-by-step

1. Build regex from the full predation phrase + keyword vocabulary.
2. Search for [source] ... [rel] ... [target] up to 180 characters apart.
3. Handle passive: [target] ... is eaten by ... [source].
4. **Category expansion fallback:** “eats insects”, expands to “eats fly”

Strength: variable-distance regex captures paraphrases missed by Method B’s fixed window.

Strength: category expansion recovers edges when Wikipedia mentions a category (“insects”) rather than a specific animal.

Limitation: generous distance windows may allow subject–relation–object triples that are not actually related.

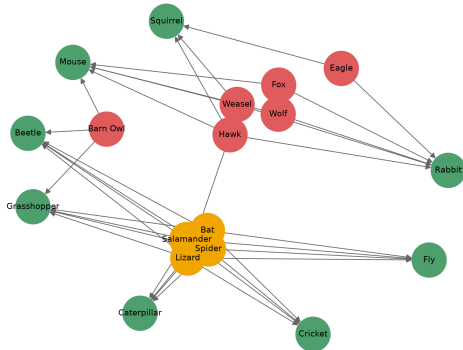
Limitation: unconstrained category expansion can introduce false positives for all members of a category.

Expected: better recall than B, moderate precision.

Method C vs Gold Standard

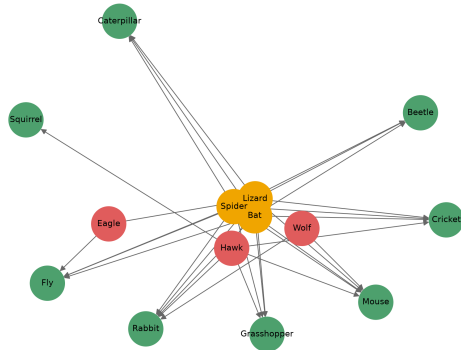
Gold standard

Gold Standard Food Web



Method C

Method C - OpenIE-style Extraction



Scores: Precision = 0.677, Recall = 0.583, F1 = 0.627 (TP = 21, FP = 10, FN = 15)

Method D — Hybrid Extraction (Pipeline)

Step 1 — Explicit verb proximity

- ▶ Tokenise sentence and detect predation verbs
- ▶ Compute token distance between verb and target animal mention
- ▶ Add edge if distance ≤ 8 tokens
- ▶ Handles lexical variants (mouse/mice, fly/flies)
- ▶ Supports passive constructions (e.g. “is eaten by”)

Step 2 — Restricted category expansion

- ▶ Triggered only if Step 1 finds no edge
- ▶ Expands only insect-related prey
- ▶ Applied only to insectivore sources: Salamander, Lizard, Spider, Bat

Method D — Strengths & Limitations

Strengths

- ▶ Strong precision via token-distance constraint
- ▶ Reduces co-occurrence noise from Method A
- ▶ Handles passive constructions reliably
- ▶ Explicit morphological matching improves recall stability
- ▶ Controlled fallback avoids OpenIE over-generation

Limitations

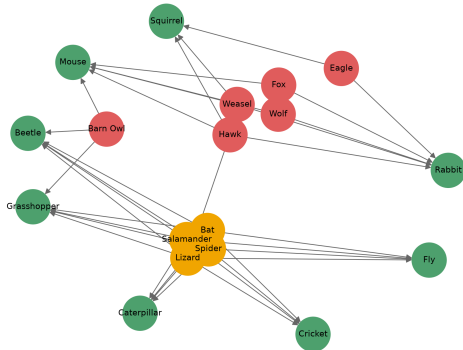
- ▶ Strict ≤ 8 token window misses long dependencies
- ▶ Mammal prey depends entirely on Step 1
- ▶ Category expansion limited (insects only)
- ▶ Still sensitive to sentence structure variation
- ▶ Cannot resolve cross-sentence relations

Overall: best trade-off between precision and recall among all methods

Method D vs Gold Standard

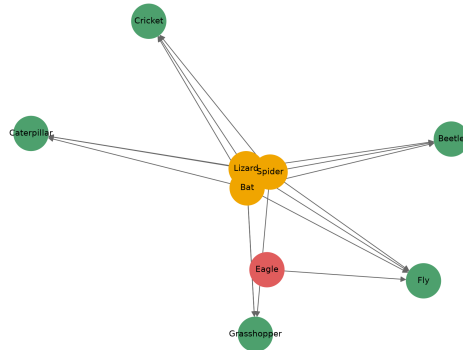
Gold standard

Gold Standard Food Web



Method D

Method D - Hybrid Extraction



Scores: Precision = 0.938, Recall = 0.417, F1 = 0.577 (TP = 15, FP = 1, FN = 21)

Method Design Comparison

	A Co-occurrence	B Verb patterns	C OpenIE	D Hybrid
Predation verb required	No	Yes	Yes	Yes
Direction inference	No	Yes	Yes	Yes
Passive voice handled	—	Yes	Yes	Yes
Category expansion	None	None	Full	Restricted
Proximity constraint	None	Window	Char. distance	Token ≤ 8
Noisy-context filter	No	No	Yes	No
Variant matching	Yes	Partial	Yes	Yes
Expected precision	Low	High	Medium	Medium-high
Expected recall	High	Medium	High	Medium-high

Evaluation Metrics

Edge-level evaluation

Let $\hat{E}$ be the set of predicted edges and E^* the gold standard.

$$\text{Precision} = \frac{|\hat{E} \cap E^*|}{|\hat{E}|} \quad \text{Recall} = \frac{|\hat{E} \cap E^*|}{|E^*|}$$

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}$$

- ▶ $|\hat{E} \cap E^*|$ = true positives (TP)
- ▶ $|\hat{E}| - \text{TP}$ = false positives (FP)
- ▶ $|E^*| - \text{TP}$ = false negatives (FN)

Scope constraint

Only edges that satisfy the trophic direction are considered valid:

- ▶ Source must be in **top predators** $\cup$ **mid-level**.
- ▶ Target must be in **mid-level** $\cup$ **prey**.
- ▶ Self-loops are never valid.
- ▶ Top predators cannot appear as prey targets.

All four methods use this same constraint, implemented via `sanitize_edges()`.

Results

Method	P	R	F1	TP	FP	FN
A — Co-occurrence	0.600	0.417	0.492	15	10	21
B — Dependency Patterns	0.625	0.139	0.227	5	3	31
C — OpenIE-style Relation Extraction	0.604	0.806	0.690	29	19	7
D — Hybrid	0.952	0.556	0.702	20	1	16

Error Analysis

why does it fail? Please finish this slide

Please finish this slide

Summary

	Method	Core signal	Best for
A	Co-occurrence	sentence co-occurrence	maximum recall
B	Verb patterns	predation verb in window	maximum precision
C	OpenIE	regex triple + category expansion	balanced recall
D	Hybrid	verb proximity ≤ 8 tokens + restricted expansion	best F1